FINE-TUNING OWSM-CTC WITH MIXTURE-OF-EXPERTS FOR EFFICIENT SPEECH RECOGNITION

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ABSTRACT

We fine-tune the OWSM-CTC v4 encoder by replacing the dense position-wise MLP layers in selected E-Branchformer blocks with Mixture-of-Experts (MoE) layers via sparse upcycling. To address the routing instability common in low-resource fine-tuning, we introduce a soft inductive router bias that guides expert specialization according to language families (e.g., Romance vs. Slavic). We evaluate our approach on LibriSpeech (100h) and the multilingual FLEURS dataset. Our results show that MoE recovers the performance of the dense fine-tuned baseline on LibriSpeech while significantly improving generalization on held-out languages in multilingual settings by mitigating negative interference. Furthermore, we analyze the accuracy–efficiency trade-off, demonstrating that optimizations such as stacked expert weights allow the model to double its parameter count (1B $\rightarrow$ 2.1B) with sublinear latency scaling, maintaining practical real-time throughput compared to the dense baseline.

Index Terms— Mixture-of-Experts (MoE), speech recognition, OWSM-CTC, sparse upcycling, multilingual ASR, expert routing, inductive bias, efficiency, FLEURS

1. INTRODUCTION

Automatic speech recognition (ASR) has evolved significantly from hybrid HMM-GMM systems to fully neural, end-to-end (E2E) architectures. Deep neural network acoustic models [1] and the Connectionist Temporal Classification (CTC) loss [2] enabled robust sequence training, while attention-based encoder-decoder models [3,4] improved the modeling of long-range dependencies. The Transformer [5] and its speech-oriented variants, such as the Conformer [6] and E-Branchformer [7], have since become the standard backbones for state-of-the-art ASR, widely adopted in toolkits like ESPnet [8].

Building on these advances, foundation models like OWSM [9] and Whisper rely on dense Transformer backbones to achieve impressive performance. However, these dense architectures fundamentally couple parameter count with computational cost: every parameter is active for every token. This creates a bottleneck for real-time inference and, in multilingual settings, leads to "negative interference," where diverse languages compete for limited capacity within shared weights [9, 10].

Mixture-of-Experts (MoE) architectures offer a promising solution by enabling conditional computation [10–12], effectively decoupling capacity from inference cost. However, naively applying MoE to pre-trained speech backbones often results in routing instability and utilization overhead.

[*] Equal contributions

In this work, we propose a MoE-based E-Branchformer architecture for the OWSM-CTC framework. We redesign the encoder by selectively replacing dense position-wise feed-forward networks (FFN) with our own MoE layers initialized via Sparse Upcycling [13]. To address the optimization difficulties inherent in fine-tuning—specifically the "cold start" problem where routers fail to specialize before the model converges—we introduce a routing strategy that combines a load-balancing auxiliary loss with a **Soft Language Prior**. This prior injects domain knowledge into the routing logic, guiding expert selection during the critical early phases of limited-data training.

Our specific contributions are as follows:

1. **Sparse Upcycling with Efficient Inference:** We initialize experts by cloning pre-trained dense weights, preserving learned representations. To mitigate the kernel overhead of MoE on high-density speech frames, we implement **expert batching** and **stacked-weight kernels**. We demonstrate on *LibriSpeech* [14] that this yields a sub-linear increase in latency ($\approx 30\%$), maintaining competitive real-time throughput against the dense baseline.

2. **Router Stabilization Strategy:** We address the router instability common in low-resource fine-tuning by combining a Switch-style **Auxiliary Loss** with a **Soft Language Prior**. This bias "nudges" the router to group linguistically similar languages (e.g., Romance vs. Slavic) early in training. This strategy prevents router collapse and enforces specialization without requiring massive datasets to learn routing patterns from scratch.

3. **Multilingual Generalization:** We extend our evaluation to the *FLEURS* multilingual dataset [15]. Experiments show that our architecturally enforced priors successfully segregate languages into distinct experts, mitigating negative interference. This results in significant performance gains on seen languages and improved zero-shot generalization on held-out languages (e.g., Slovak, Croatian).

2. PROBLEM FORMULATION

2.1. Setting.

Let $\mathbf{x} = (x_1, \ldots, x_T)$ be acoustic frames with $x_t \in \mathbb{R}^{d_x}$. An encoder produces hidden states $\mathbf{H} = (h_1, \ldots, h_T)$, $h_t \in \mathbb{R}^d$, which are mapped to framewise logits for CTC. Training minimizes the negative log-likelihood $\mathcal{L}_{\mathrm{ASR}}(\theta)$, where the CTC likelihood sums over all valid alignment paths.

2.2. Dense E-Branchformer block (baseline).

Given $\mathbf{H}^{\ell-1} \in \mathbb{R}^{T\times d}$, layer ℓ applies a macaron (pre) FFN, a dual branch (self-attention and CGMLP with fusion), and a post FFN:

$$\tilde{\mathbf{H}}_0^\ell = \mathbf{H}^{\ell-1} + s\,\mathrm{FFN}_{\mathrm{pre}}^\ell(\mathbf{H}^{\ell-1}), \tag{1}$$

$$\tilde{\mathbf{H}}_1^\ell = \tilde{\mathbf{H}}_0^\ell + \mathcal{B}^\ell(\tilde{\mathbf{H}}_0^\ell), \tag{2}$$

$$\mathbf{H}^\ell = \tilde{\mathbf{H}}_1^\ell + s\,\mathrm{FFN}_{\mathrm{post}}^\ell(\tilde{\mathbf{H}}_1^\ell), \tag{3}$$

where $s \in (0, 1]$ is the macaron scale (typically $s = \frac{1}{2}$), $\mathcal{B}^\ell(\cdot)$ is the attention+CGMLP branch with fusion, and each dense positionwise FFN acts independently per time step:

$$\mathrm{FFN}(h) \;=\; W_2\,\sigma(W_1 h) \;\in \mathbb{R}^d, W_1 \in \mathbb{R}^{d_{\mathrm{ff}}\times d}, W_2 \in \mathbb{R}^{d\times d_{\mathrm{ff}}}. \tag{4}$$

2.3. MoE-Enhanced Layer

We modify selected encoder layers $\ell \in \mathcal{I}$ by replacing FFN's (Eq. 1, 3) with a Mixture-of-Experts (MoE) module.

Router with Inductive Bias: For a token h, let f be the language family index. The router computes logits z using a standard learnable affine transformation plus a **static** inductive bias:

$$z = (W_r h + b_r) + b_{\mathrm{prior}}[f] + \varepsilon, \tag{5a}$$

$$p = \mathrm{softmax}(z) \in \mathbb{R}^E, \tag{5b}$$

$$\varepsilon \sim \mathcal{N}(\mathbf{0},\, \sigma_g^2 I_E) \quad \text{(train only)}, \tag{5c}$$

where $b_{\mathrm{prior}} \in \mathbb{R}^{F\times E}$ is a fixed, small non-learnable buffer initialized to encourage grouping of linguistically similar languages. We explicitly design this bias to be static because, in limited-data regimes, a learnable prior risks converging to a degenerate solution (learning a 'bad prior') before the router receives sufficient gradient signal. Conversely, in scenarios where we are not constrained by data availability, this static bias would be omitted to allow the router to naturally learn data-driven specializations.

Token Dispatch: We employ Top-K routing. Let $\mathcal{S}_K(h)$ be the set of K expert indices with the highest probabilities in p. Experts are parameterized as stacked tensors $\mathcal{W}_1, \mathcal{W}_2$ to allow batched execution. The selected gates are re-normalized: $\tilde{p}_e = p_e / \sum_{j\in\mathcal{S}_K} p_j$. Let $m_{h,e} \in \{0, 1\}$ indicate if token h is kept after capacity enforcement. The output is:

$$\mathrm{MoE}(h) \;=\; \sum_{e\in\mathcal{S}_K(h)} m_{h,e}\,\tilde{p}_e \cdot (\mathcal{W}_2[e]\,\sigma(\mathcal{W}_1[e]\,h))\,. \tag{6}$$

During training, tokens exceeding the capacity limit $C = \lceil \phi N/E \rceil$ are dropped ($m_{h,e} = 0$), with the residual connection passing information forward.

2.4. Auxiliary Load Balancing.

We optimize $\mathcal{L} = \mathcal{L}_{\mathrm{ASR}} + \lambda\mathcal{L}_{\mathrm{bal}}$. Following the Switch Transformer [11], the auxiliary loss penalizes the dot product of the routing fraction (importance) and the dispatch fraction (load):

$$\mathcal{L}_{\mathrm{bal}} \;=\; E\sum_{e=1}^{E}\left(\frac{1}{N}\sum_{i=1}^{N} p_{i,e}\right)\cdot\left(\frac{1}{N}\sum_{i=1}^{N} m_{i,e}\right) \tag{7}$$

where $m_{i,e}$ is the binary mask indicating if token i was routed to expert e.

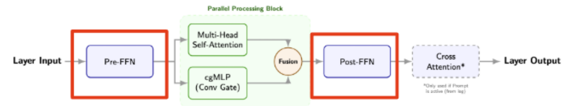


Fig. 1: Overview of the OWSM-CTC E-Branchformer block. In our sparse upcycling setup, selected dense FFN layers are replaced by MoE-FFNs with a language-prior-guided router.

3. METHOD

In this section, we describe the specific instantiation of the MoE-enhanced OWSM system, our sparse upcycling strategy, and the inference optimizations employed to mitigate computational overhead.

3.1. Baseline: OWSM-CTC v4

Our baseline is the OWSM-CTC v4 Medium model [9], a fully open-source speech foundation model trained on public data. The encoder consists of $L = 27$ stacked E-Branchformer blocks with a model dimension of $d = 1024$. Each block contains a standard attention module and a dense position-wise FFN. This baseline contains approximately 1 billion parameters.

3.2. MoE Upcycling and Architecture

Instead of training a sparse model from scratch, we employ Sparse Upcycling [13]. We initialize the MoE layers by copying the weights from the pre-trained dense FFNs into each expert. This allows the model to retain the acoustic capabilities learned during the massive pre-training phase while expanding its capacity for specialization.

Layer Selection and Expert Configuration: We replace the dense FFNs with MoE layers in a subset of encoder layers denoted by $\mathcal{I}_{MoE}$. As shown in Figure 1, we replace the dense FFN in selected E-Branchformer blocks with a sparse MoE-FFN, while keeping the original convolutional and self-attention components untouched. The number of experts E and the selected layers $\mathcal{I}_{MoE}$ are hyperparameters adapted to the task:

- For **single-language validation**, we maximize capacity by applying MoE to the upper 9 layers ($l \in [18, 26]$) with $E = 8$ experts.
- For **multilingual fine-tuning**, to mitigate overfitting on smaller datasets while retaining specialization capabilities, we target the last 4 layers ($l \in [23, 26]$) with $E = 4$ experts.

In all configurations, we use Top-$K = 1$ routing to keep the active parameter count per token identical to the dense baseline.

Implementation of Language Prior: As formulated in Section 2.3, we inject a static bias b_{prior} into the router. We map each target language to a language family ID (e.g., Spanish/Portuguese $\to$ Romance, Polish/Czech $\to$ Slavic). The bias vector is manually initialized to encourage the router to prefer specific subsets of experts for each family during the initial training phase, facilitating the emergence of language-specialized experts.

3.3. Inference Optimization

A naive MoE implementation often suffers from memory fragmentation and kernel launch overhead due to the scattering of tokens. To address this, we implement Expert-wise Token Batching:

- We first compute routing decisions for all tokens in a batch.

Table 1: Datasets used in each experiment.

Experiment	Dataset and size
1. Single-language validation	LibriSpeech train-clean-100 (100 h, ∼28.5k training utterances). Validation on `dev-clean`/`dev-other` (∼2.7k / 2.8k utterances) and evaluation on `test-clean`/`test-other` (∼2.6k / 2.9k utterances).
2. Multilingual tests	FLEURS multilingual subset. 4 languages: Spanish, Portuguese, Polish, and Czech, with about 2.5k training utterances per language. Validation and test sets use the same 4 languages with similar sizes. For zero-shot evaluation we additionally use Italian, French, Slovak, and Croatian.
3. Performance analysis	LibriSpeech `test-clean` and `test-other` evaluation sets, decoded with each model to measure runtime performance.

- We explicitly group tokens assigned to the same expert into contiguous memory blocks.
- We perform batched matrix multiplications (GEMM) for each expert in parallel, rather than processing tokens individually.

This optimization significantly improves GPU utilization and reduces the inference latency gap between the dense and MoE models, as demonstrated in our experiments.

3.4. Training Setup

We fine-tune using AdamW with learning rate 3×10^{-5}, weight decay 0.01, and $\beta = (0.9, 0.98)$. We use an effective batch size of 16, gradient clipping at 5.0, and mixed-precision training. SpecAugment is applied with frequency and time masking. For MoE layers, we set noisy gating $\sigma_g = 1.5$, capacity factor $\phi = 1.25$, and auxiliary loss weight $\lambda = 0.01$. Expert weights are initialized from the pretrained FFN with small Gaussian noise ($\sigma_{\text{init}} = 0.01$) to encourage differentiation. The language-family bias b_{prior} remains fixed (non-learnable) throughout training.

4. EXPERIMENTS

4.1. Datasets

We conducted experiments on two datasets to validate efficiency and multilingual generalization capabilities.

- **LibriSpeech**: We utilized the *train-clean-100* subset (100 hours) for single-language fine-tuning. Evaluation was performed on the standard *test-clean* and *test-other* splits.
- **FLEURS**: To investigate cross-lingual transfer, we selected four languages from the FLEURS dataset: Spanish and Portuguese (Romance family), and Polish and Czech (Slavic family). We evaluated performance on these seen languages and four **held-out** languages—Italian and French (Romance), Slovak and Croatian (Slavic)—to test zero-shot generalization within each language family.

Preprocessing: All audio was resampled to 16 kHz. We extracted 80-dimensional log-Mel filterbank features using a 25ms window and 10ms shift, followed by global mean-variance normalization. Text was tokenized using the pre-trained OWSM BPE tokenizer (based on SentencePiece [16]) to ensure vocabulary consistency.

4.2. Baseline

Our baseline is the **OWSM-CTC v4 Medium** model, a 1B-parameter encoder-only architecture. The encoder consists of 27 E-Branchformer blocks. For fair comparison, we fine-tuned this dense baseline on the target datasets using optimization settings identical to our MoE variants, serving as the "Dense Fine-Tuned" reference.

4.3. Experimental Setups

We implemented two distinct MoE configurations tailored to the specific challenges of each experiment, utilizing sparse upcycling from the dense checkpoint.

1. Single-Language Configuration (LibriSpeech):

- **Architecture:** We replaced the dense FFNs with MoE layers in the **last 9 encoder layers** (layers 18–26).
- **Experts:** We employed $E = 8$ experts with Top-$K = 1$ routing.
- **Hardware:** Fine-tuning and evaluation were performed on a single NVIDIA A100 (40GB) GPU.

2. Multilingual Configuration (FLEURS):

- **Architecture:** To manage the complexity of multilingual representations, we applied MoE replacement to the **last 4 encoder layers** (layers 23–26).
- **Experts:** We utilized $E = 4$ experts with Top-$K = 1$ routing.
- **Router Bias:** We enabled the Soft Language Prior (as defined in Section 3) to guide experts towards language-family specialization.
- **Hardware:** Experiments were conducted on an NVIDIA A100 (80GB) GPU.

Training Details: We optimized the models using AdamW [17] with mixed precision. The auxiliary load-balancing coefficient was set to $\lambda = 0.01$.

4.4. Results and Discussion

As shown in Figure 2, the auxiliary load-balancing loss rises quickly during the early phase of training, driven by the uneven routing patterns. Once the global entropy ratio and expert loads stabilize, the auxiliary loss also flattens out, while the main CTC loss keeps decreasing. This suggests that the language-prior-guided router successfully avoids expert collapse without hurting the primary ASR objective.

Experiment 1: Efficiency Validation on LibriSpeech Table 2 compares the efficiency and accuracy of the MoE model ($E = 8, L_{18-26}$) against the dense baseline. The MoE model achieved a WER of **5.07%** on *test-other*, slightly surpassing the fine-tuned dense baseline (5.15%). Crucially, despite doubling the parameter count to 2.07B, the inference latency increased by only $\approx$ 31% (42.1ms vs. 55.1ms), confirming that sparse upcycling effectively decouples model capacity from computational cost.

Experiment 2: Multilingual Specialization on FLEURS Table 3 presents the results of the 4-expert configuration (L_{23-26}) on the multilingual task. The MoE model demonstrated superior performance across all metrics. Most notably, we observed significant gains in zero-shot generalization to held-out languages. For Slovak (unseen Slavic), the MoE model reduced WER from 40.01%

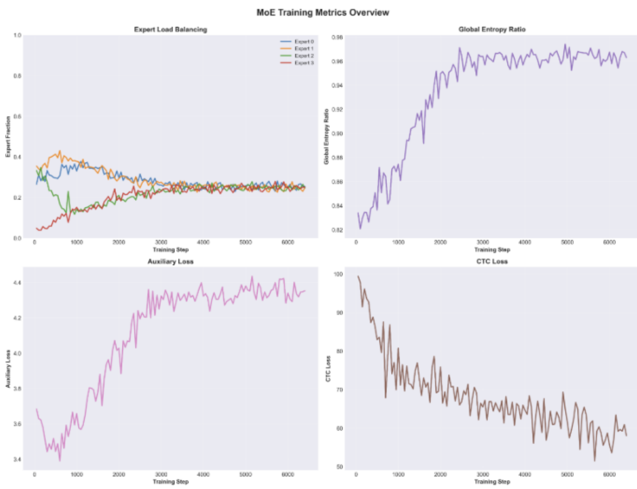


Fig. 2: MoE training dynamics on LibriSpeech showing expert load, entropy, auxiliary loss, and CTC loss. The auxiliary loss (bottom left) rises early to penalize the imbalance favored by the greedy CTC objective (bottom right), eventually stabilizing as the router balances acoustic accuracy with expert capacity.

Table 2: Comparison of WER (%) and Efficiency on LibriSpeech (100h). Metrics measured on a single A100 40GB GPU.

Model	**Test-Clean** WER (%)	**Test-Other** WER (%)	**Latency** (ms/utt)	**Throughput** (utt/s)	**Params** (B)
Dense (Baseline)	3.26	6.59	46.4	21.2	1.0
Dense (Fine-Tuned)	**2.36**	5.15	**42.1**	**24.0**	1.0
MoE (Fine-Tuned)	2.39	**5.07**	55.1	19.8	2.07

to 30.02%, a relative improvement of 25%. Similarly, Croatian improved by 21.6%. This indicates that the language-prior guided routing successfully captured language-family-invariant features (e.g., Slavic acoustic patterns) in the upper layers of the encoder.

As shown in Figure 3a and 3b, the router indeed specializes different experts for Romance and Slavic language families, and this pattern remains stable across the top encoder layers.

5. CONCLUSION

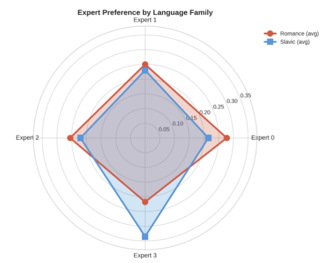
In this work, we explored the application of Mixture-of-Experts (MoE) to the open-weight OWSM-CTC v4 speech foundation model via sparse upcycling. Our investigation focused on two key aspects: the efficiency-accuracy trade-off in single-language ASR and the potential for expert specialization in multilingual settings.

First, our experiments on LibriSpeech validated that replacing dense FFNs with MoE layers is a practical strategy for scaling ASR models. We demonstrated that the MoE architecture effectively doubled the parameter capacity (from 1B to 2.07B) while incurring only a moderate 31% increase in inference latency. This confirms that conditional computation allows for substantial capacity expansion with sub-linear computational costs.

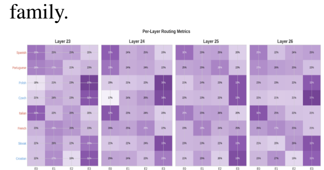
Second, our multilingual experiments on FLEURS highlighted the effectiveness of our novel **Soft Language Prior Routing**. By injecting a linguistic inductive bias into the router, we successfully guided experts to specialize in distinct language families (Romance vs. Slavic). This specialization led to dramatic improvements in zero-shot generalization, reducing WER by up to 25% on held-out languages such as Slovak. These findings suggest that MoE, when combined with linguistically informed routing, offers a powerful solution to the interference trade-offs inherent in dense multilingual models.

Table 3: Multilingual WER (%) on FLEURS (MoE Config: 4 Experts, Layers 23-26). The MoE model shows massive gains on Held-Out languages.

Type	**Language**	**WER (%)** Dense FT	MoE FT	**Relative Change**
Seen	Spanish	5.12	**4.89**	-4.5%
	Portuguese	6.75	**6.57**	-2.7%
	Polish	21.21	**20.65**	-2.6%
	Czech	27.52	**25.39**	-7.7%
Held-Out	Italian	6.61	**5.64**	-14.7%
	French	11.90	**10.75**	-9.7%
	Slovak	40.01	**30.02**	**-25.0%**
	Croatian	51.53	**40.39**	**-21.6%**



(a) Expert preference by language family.



(b) Per-layer routing heatmap (layers 23–26).

Fig. 3: Expert specialization induced by the language-prior-guided router. (a) Average expert preference for Romance vs. Slavic languages on FLEURS. (b) Per-layer routing heatmap, showing that the same experts consistently specialize in particular language families across the top Transformer layers.

6. DISCUSSION WITH INSTRUCTION/TA

Yes. We have a discussion with Masao Someki all the content above.

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